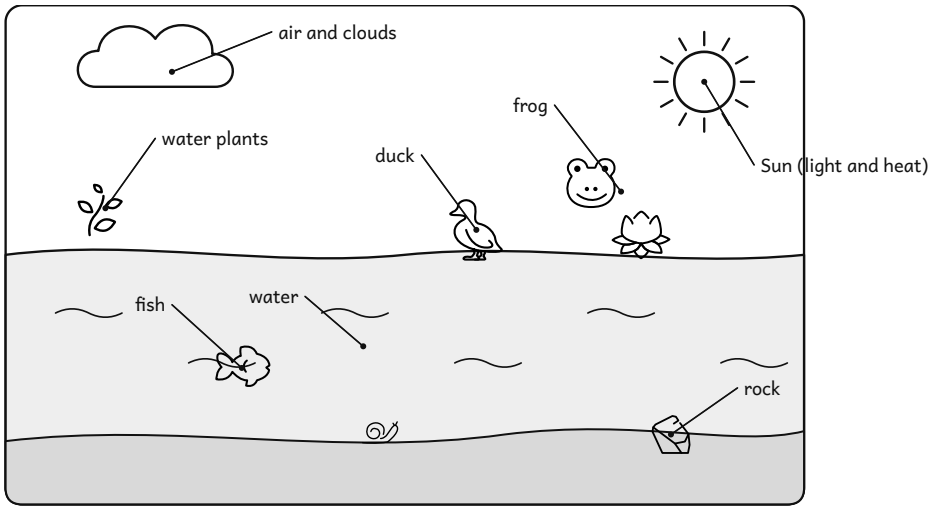
LIVING THINGS

# Ecosystems and Interactions

Living and non-living parts, interactions, levels from cell to biosphere, and ecosystems as systems




**In this pack**

**Learn it:** Ecosystems: everything is connected ★☆☆
**Fact check:** Parts of an ecosystem ★☆☆
**Think it through:** When something changes ★★☆
**Investigate:** Do snails prefer damp or dry places? ★★☆
**Real-world case study:** Rice and fish: farming partners ★★★
**Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science Stage 4: Unit 2 Living things and environments (2.2 A habitat for snails).

**This pack belongs to:** _____ **Class:** _____

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# Ecosystems: everything is connected

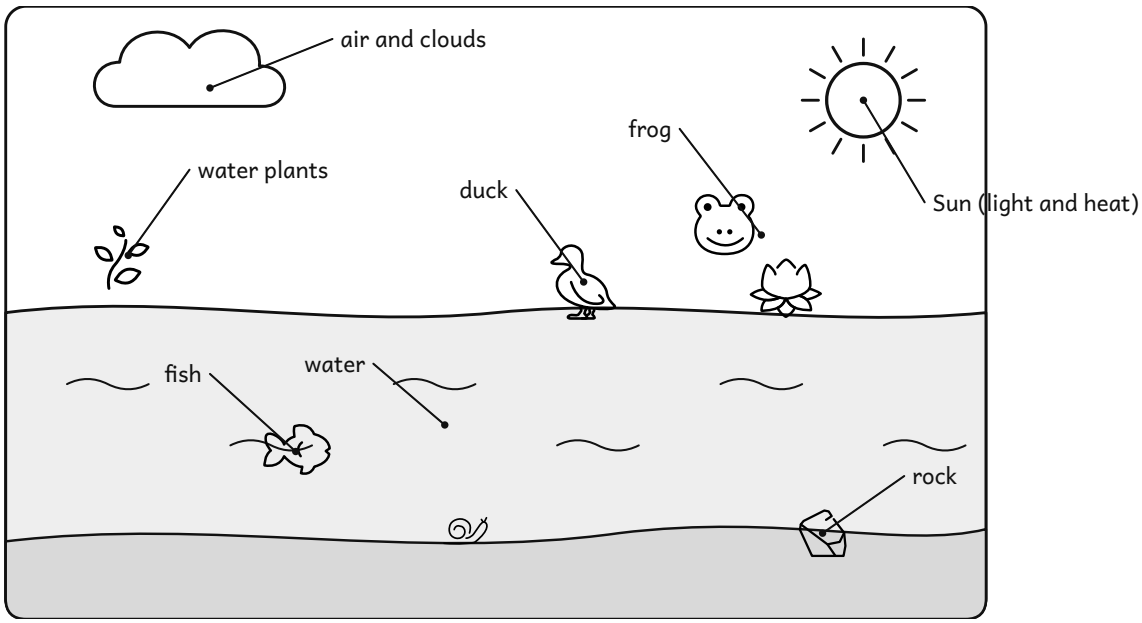
★☆☆
Grade 4

Name: Class: Date:

**Learning goals**

- Explain what an ecosystem is and sort its parts into living and non-living.
- Give examples of interactions in an ecosystem and explain them.
- Describe the levels from a single cell up to the whole biosphere.

Sit quietly beside a village pond. Fish dart under the water, a frog sits on a lily pad, a duck dabbles for snails, and the Sun warms the water. All these things, living and non-living, work together. Together they make an **ecosystem**.




*A pond ecosystem.*

## What is an ecosystem?

An **ecosystem** is all the **living things** in a place together with the **non-living things** around them, and all the ways they affect each other. A pond, a rice field, a forest, a haor and even a rotting log are ecosystems.

| Living (biotic) | Non-living (abiotic) |
|---|---|
| plants, animals, fungi, tiny microbes | sunlight, water, air, soil, rocks, temperature |

How can you tell if something is living? Living things **move**, **grow**, **feed**, **breathe**, **get rid of waste**, **sense** things around them and **reproduce** (have young). Non-living things do not do all of these.

### Living things and non-living things

- Plants use **sunlight**, **water** and **air** to make their food.
- Fish take in **oxygen** that is dissolved in the **water**.
- Earthworms mix the **soil** and make it better for roots.
- Trees give **shade**, which keeps the ground cooler and damper.

### Living things and other living things

- **Feeding:** a kingfisher eats fish; a cow eats grass.
- **Competing:** two plants grow side by side and compete for light and water.
- **Helping each other:** bees get nectar from flowers and carry pollen for them.
- **Shelter:** birds nest in trees; crabs hide in mangrove roots.

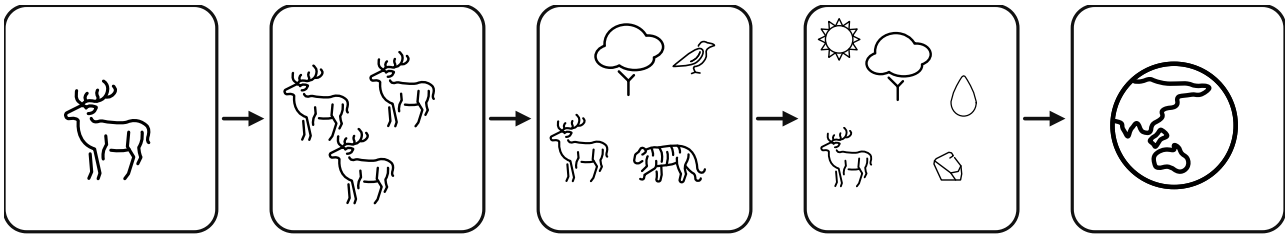
**Think like a scientist**

Why do these interactions happen? Every living thing needs **food**, **water**, **shelter**, **light** (for plants) and a place to **raise young**. Interactions are how they get them.

### From cells to the whole Earth

Living things are made of tiny **cells**. Cells work together as **tissues**, tissues make **organs** (like the heart), and organs work together in **organ systems**. A whole living thing is an **organism**.

Living things also form bigger groups: all the deer of one kind in a forest are a **population**; all the different populations living together are a **community**; the community plus the non-living things is an **ecosystem**; and all the ecosystems on Earth make the **biosphere**.




*Bigger and bigger levels.*

### Ecosystems are systems

A **system** is a group of parts that work together. If one part changes, the others change too. If a pond dries up, the fish die, the kingfishers must fly away, and the frogs cannot lay their eggs. Studying ecosystems as systems helps scientists predict what will happen when something changes.

**Bangladesh connection**

Bangladesh has many kinds of ecosystem: mangrove forest in the Sundarbans, hill forests in Chattogram, haors and beels in the north-east, coral around Saint Martin's Island, rivers, rice fields and city parks.

**Try this at home**

Lift a stone or a log in a garden. Write a list of the living things and the non-living things you find. Put it back gently afterwards!

**Key words**

| | |
|---|---|
| **ecosystem** | living and non-living things in a place, and how they affect each other |
| **interaction** | when one thing affects another |
| **compete** | try to get the same thing as another living thing |
| **population** | all the living things of one kind in a place |
| **community** | all the populations living together in a place |
| **biosphere** | all the ecosystems on Earth |
| **organism** | a whole living thing |
| **system** | parts that work together |

FACT CHECK · A4 · ECOSYSTEMS

# Parts of an ecosystem

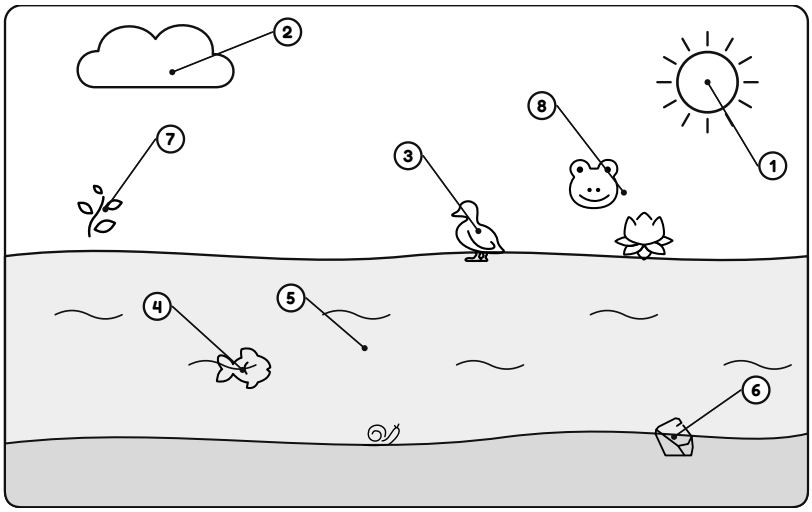
★☆☆
Grade 4

Name: Class: Date:

**Learning goals**

- Sort living and non-living parts.
- Order the levels of life.

**1** **Look at the pond. Write the name of each numbered part and say if it is living or non-living.** [8]




| Number | Name | Living or non-living? |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |
| 8 | | |

**2** **Put these levels in order, from smallest to biggest.** [4]

- ☐ ecosystem
- ☐ biosphere
- ☐ organ
- ☐ population
- ☐ cell
- ☐ organism
- ☐ tissue
- ☐ community

**3** **Match each interaction to its type.** [4]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| a heron eats a frog ● | ● **A** feeding |
| two trees grow towards the same patch of light ● | ● **B** helping each other |
| a bee carries pollen while it drinks nectar ● | ● **C** competing |
| a crab hides between mangrove roots ● | ● **D** shelter |

THINK IT THROUGH · A4 · ECOSYSTEMS

# When something changes

★★☆
Grade 4

Name: Class: Date:

**Learning goals**

- Explain interactions.
- Predict the effects of change.

**1** **Explain how each non-living thing helps the living things in a pond.** [3]

| Non-living thing | How it helps |
|---|---|
| sunlight | |
| water | |
| mud at the bottom | |

**2** **A factory pours hot water into a pond. The water gets much warmer and holds less oxygen. Predict what happens to:** [3]

**a)** the fish

**b)** the kingfishers that eat the fish

**c)** the people who fish in the pond

**3** **Is a pond a system? Explain using the word parts.** [2]

**4** **Rafi found a dry seed. It does not move or grow. Is it living or non-living? Explain.** [2]

# Do snails prefer damp or dry places?

★★☆ Grade 4

Name: Class: Date:

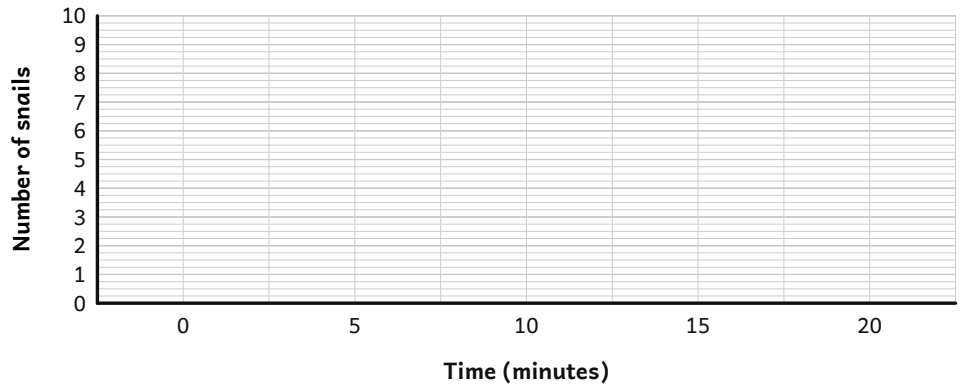
**Learning goals**

- Use a choice chamber.
- Record results over time.
- Explain results using habitats.

Class 4D put 10 garden snails in the middle of a tray. One half of the tray had damp paper towel; the other half had dry paper towel. Every 5 minutes they counted the snails on each side. Then they put the snails back in the garden.

**1** **Here are the results.** [5]

| Time (minutes) | 0 | 5 | 10 | 15 | 20 |
|---|---|---|---|---|---|
| Snails on the damp side | 0 | 3 | 6 | 8 | 9 |
| Snails on the dry side | 0 | 2 | 2 | 1 | 1 |




**a)** Plot both lines on the graph.

**b)** Describe what happened.

**2** **Why do snails prefer damp places?** [2]

**3** **Why did they start with all the snails in the middle?** [1]

**4** **How could they make the results more reliable?** [1]

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REAL-WORLD CASE STUDY · A4 · ECOSYSTEMS

# Rice and fish: farming partners

★★★
Grade 4

**Name:** **Class:** **Date:**

**Learning goals**

- Find interactions in a farm ecosystem.
- Use data to compare farms.
- Explain the benefits for people and nature.

Science · Farming · Maths · Economics

In many parts of Bangladesh, farmers keep **fish in their rice fields**. This is called **rice-fish farming**. The field is flooded with water, the rice plants grow, and small fish swim between them.

The rice and the fish help each other. The fish eat insects and weeds that would harm the rice, so the farmer uses less **pesticide**. Fish droppings act as **fertiliser** for the rice. The rice plants give the fish **shade** and shelter. At the end of the season the farmer harvests both rice and fish, which means more food and more money.

Based on research on integrated rice-fish farming in Bangladesh (WorldFish and the Bangladesh Rice Research Institute). Numbers below are example data.

**1** **Find the interactions in the rice-fish field.** [3]

| Who helps? | How? |
|---|---|
| fish help rice | |
| rice helps fish | |
| water helps both | |

**2** **Two farms (example data).** [3]

| Farm | Rice harvested (kg) | Fish harvested (kg) | Pesticide used (litres) |
|---|---|---|---|
| rice only | 500 | 0 | 6 |
| rice and fish | 520 | 60 | 2 |

**a)** How much more rice did the rice-fish farm harvest?

.....

**b)** Why did the rice-fish farm use less pesticide?

.....

**3** **Why is using less pesticide good for the ecosystem?** [2]

.....

.....

**4** **Draw a diagram showing three interactions in a rice-fish field. Use arrows and labels.** [3]

Rice-fish field

# Show what you know

★★★ Grade 4

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

## 1 Find the key words.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| R | T | B | E | A | H | D | V | H | K | B |
| C | D | V | P | N | P | H | A | O | Y | A |
| P | E | A | O | Y | G | H | O | R | L | T |
| M | A | L | I | E | S | P | Y | G | I | W |
| S | U | N | L | I | G | H | T | A | V | A |
| I | C | W | O | S | T | L | L | N | I | T |
| N | P | S | Y | S | T | E | M | W | N | E |
| Y | N | R | L | N | B | N | E | O | G | R |
| W | F | C | O | M | M | U | N | I | T | Y |
| U | L | O | K | F | Y | S | O | I | L | N |
| T | W | E | C | O | S | Y | S | T | E | M |

**cell community ecosystem living organ soil sunlight system water**

## 2 Which is a non-living part of an ecosystem? [1]

- ◯ frog
- ◯ rock
- ◯ grass
- ◯ mushroom

## 3 All the hilsa fish in the Meghna river are a... [1]

- ◯ community
- ◯ population
- ◯ biosphere
- ◯ organ

## 4 Draw an ecosystem near your home or school. Label two living and two non-living things, and draw one interaction arrow. [4]

My ecosystem

**5** **Challenge. Explain why scientists say "you cannot change just one thing in an ecosystem".** [2]

.....

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| explain what an ecosystem is | ○ | ○ | ○ |
| sort living and non-living parts | ○ | ○ | ○ |
| give examples of interactions and explain them | ○ | ○ | ○ |
| put the levels from cell to biosphere in order | ○ | ○ | ○ |
| investigate what conditions snails prefer | ○ | ○ | ○ |
| explain how rice and fish help each other | ○ | ○ | ○ |